

ATLANTIC COMPUTATIONAL EXCELLENCE NETWORK

MPI Programming on ACEnet

What is ACEnet?

- Shared, publicly-funded resource...
- ...for research computing...
 - physics, chemistry, oceanography, biology, math, engineering, CS, ...
- ...across Atlantic Canada
 - six clusters, plus software
 - ~11 technical personnel
 - ~600 user accounts
- See <http://www.ace-net.ca/wiki/ACEnet>

CPU counts, and memory

Cluster	Nodes	Cores/node	RAM/node	Total cores
Brasdor	269	4	8G	1152
Courtenay	13	4	16G	52
Fundy	38	16	64G, 128G	636
	7	4	16G	
Glooscap	55	4	4G	2252
	127	16	32G	
Mahone	138	4	16G, 64G	552
Placentia	154	16	32G, 64G	2884
	17	12	24G	
	54	4	16G, 64G	
	117	owned by individual groups		908

A variety of host architectures

- SunFire X4100 & X2200
 - 2x dual-core Opterons → 4-way
 - HP nodes @glooscap.ace-net.ca similar
- SunFire X4600
 - 8x dual-core Opterons → 16-way
- X6440 “Blades”
 - 4x quad-core Opterons → 16-way
 - no local disk, just small flash memory

See http://www.ace-net.ca/wiki/Compute_Resources for more

File systems

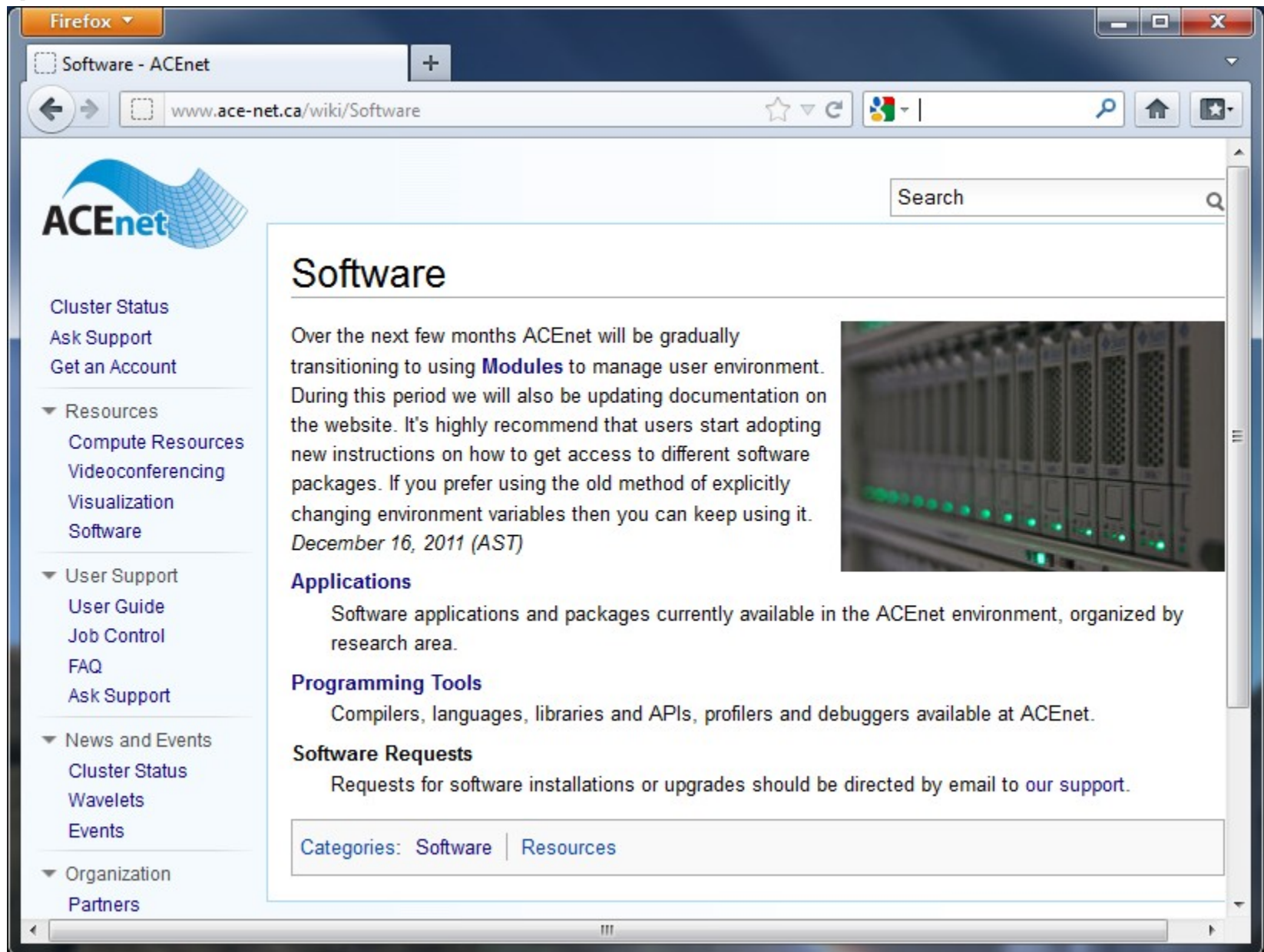
- Each cluster has own storage array
- Move data explicitly between clusters
 - with sftp or scp
- Filesystems:
 - /home NFS, individual, small DAU
 - /globalscratch NFS, individual, large DAU
 - /scratch/tmp node-local, various sizes
 - /nqs NFS, common, multi-Tb,
must be requested

See http://www.ace-net.ca/wiki/Storage_System

Storage Configurations

Cluster	/home user quota	/globalscratch user quota	/nqs size	/scratch/tmp
Brasdor	47 Gb	238 Gb	14 Tb	20 - 330 Gb
Courtenay	--	--	2.0 Tb	100 Gb
Fundy	47 Gb	238 Gb	14 Tb	35 - 100 Gb
Glooscap	61 Gb	<i>n/a</i>	<i>tbd</i>	0 - 15 Gb
Mahone	13 Gb	238 Gb	14 Tb	50 - 100 Gb
Placentia	13 Gb	238 Gb	14 Tb	0 - 140 Gb
Cluster-local				Node-local

<http://www.ace-net.ca/wiki/Software>



The screenshot shows a Firefox browser window with the address bar displaying www.ace-net.ca/wiki/Software. The page features the ACEnet logo on the left, a search bar at the top right, and a main content area with the title "Software". The left sidebar contains a navigation menu with links to Cluster Status, Ask Support, Get an Account, Resources, User Support, News and Events, and Organization. The main content area includes a paragraph about transitioning to Modules, a date "December 16, 2011 (AST)", and sections for Applications, Programming Tools, and Software Requests. A photograph of server racks is visible on the right side of the main content area. At the bottom, there are category links for Software and Resources.

Firefox

Software - ACEnet

[www.ace-net.ca/wiki/Software](#)

Search

ACEnet

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Resources

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- Software

User Support

- User Guide
- Job Control
- FAQ
- Ask Support

News and Events

- Cluster Status
- Wavelets
- Events

Organization

- Partners

Software

Over the next few months ACEnet will be gradually transitioning to using **Modules** to manage user environment. During this period we will also be updating documentation on the website. It's highly recommend that users start adopting new instructions on how to get access to different software packages. If you prefer using the old method of explicitly changing environment variables then you can keep using it.
December 16, 2011 (AST)

Applications

Software applications and packages currently available in the ACEnet environment, organized by research area.

Programming Tools

Compilers, languages, libraries and APIs, profilers and debuggers available at ACEnet.

Software Requests

Requests for software installations or upgrades should be directed by email to [our support](#).

Categories: [Software](#) | [Resources](#)

Software Development

http://www.ace-net.ca/wiki/Programming_Tools

<u>Languages</u>	<u>Compiler Suites</u>	<u>Parallel APIs</u>	<u>Libraries</u>	<u>Tools</u>
C/C++			LAPACK	TotalView
Fortran	PGI	Open MPI	FFTW	dbx, gdb
Perl	Intel	OpenMP	GSL	Valgrind
Python	Gnu	...	ACML	make
Java	SunStudio		netCDF	cvs, svn
...			HDF	git
			Boost	...
			...	

Software requests to:

support@ace-net.ca

Getting An Account

- http://www.ace-net.ca/wiki/Get_an_Account
- Register 1st with Compute Canada
 - You will need your professor's Compute Canada Role Identifier (CCRI)
- Once you are accepted at Compute Canada, request an ACEnet login:
<https://ccdb.computecanada.org/me/facilities>

Getting Connected



- Logging in
 - ssh *Secure Shell*
- Moving Data Around
 - sftp *Secure File Transfer Protocol*
 - scp *Secure Copy*
- all bundled with **Mac OS X** or **Linux**
 - ssh -X username@mahone.ace-net.ca
 - sftp username@mahone.ace-net.ca
- **Windows?** Use PuTTY and WinSCP
and Xming ... or Cygwin/X
just google for them



Command-Line Interface



```
glooscap-head.cs.dal.ca - PuTTY
Using username "rdickson".
rdickson@glooscap.ace-net.ca's password:
Last login: Wed Mar 16 10:03:16 2011 from rdickson.ucis.dal.ca
Welcome to Glooscap.ace-net.ca - Halifax, Nova Scotia
*****

rdickson@glooscap: ~ $ /usr/local/R/bin/R

R version 2.8.0 (2008-10-20)
Copyright (C) 2008 The R Foundation for Statistical Computing
ISBN 3-900051-07-0

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

    Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> █
```

Environment modules

- Manipulate \$PATH and other env. vars easily
-

```
$ module list
```

```
Currently Loaded Modulefiles:
```

- | | |
|-----------------|----------------------|
| 1) pgi/8.0-6 | 3) openmpi/pgi/1.2.9 |
| 2) sunstudio/12 | 4) totalview/8.8.0-2 |

```
$ module avail
```

```
...huge list...
```

```
$ module unload openmpi
```

```
$ module load intel
```

```
$ module help
```

```
...et cetera...
```

Parallel programming

- Shared-memory programming
 - Compilers support OpenMP (except Gnu v3)
 - "Open Multi-Processing"
- Distributed-memory programming
 - Use MPI "Message Passing Interface"
 - implementation is Open MPI
 - don't be confused by similarity in names!

MPI at ACEnet

- Open MPI v.1.2.9
- references PGI compilers
 - “mpicc” is wrapper for pgcc
 - “mpic++” wraps pgCC
 - “mpif90” wraps pgf90
- **man mpicc; man pgcc**
- see http://www.ace-net.ca/wiki/Open_MPI

MPI 'Hello, world'

```
#include <stdio.h>
#include <mpi.h>

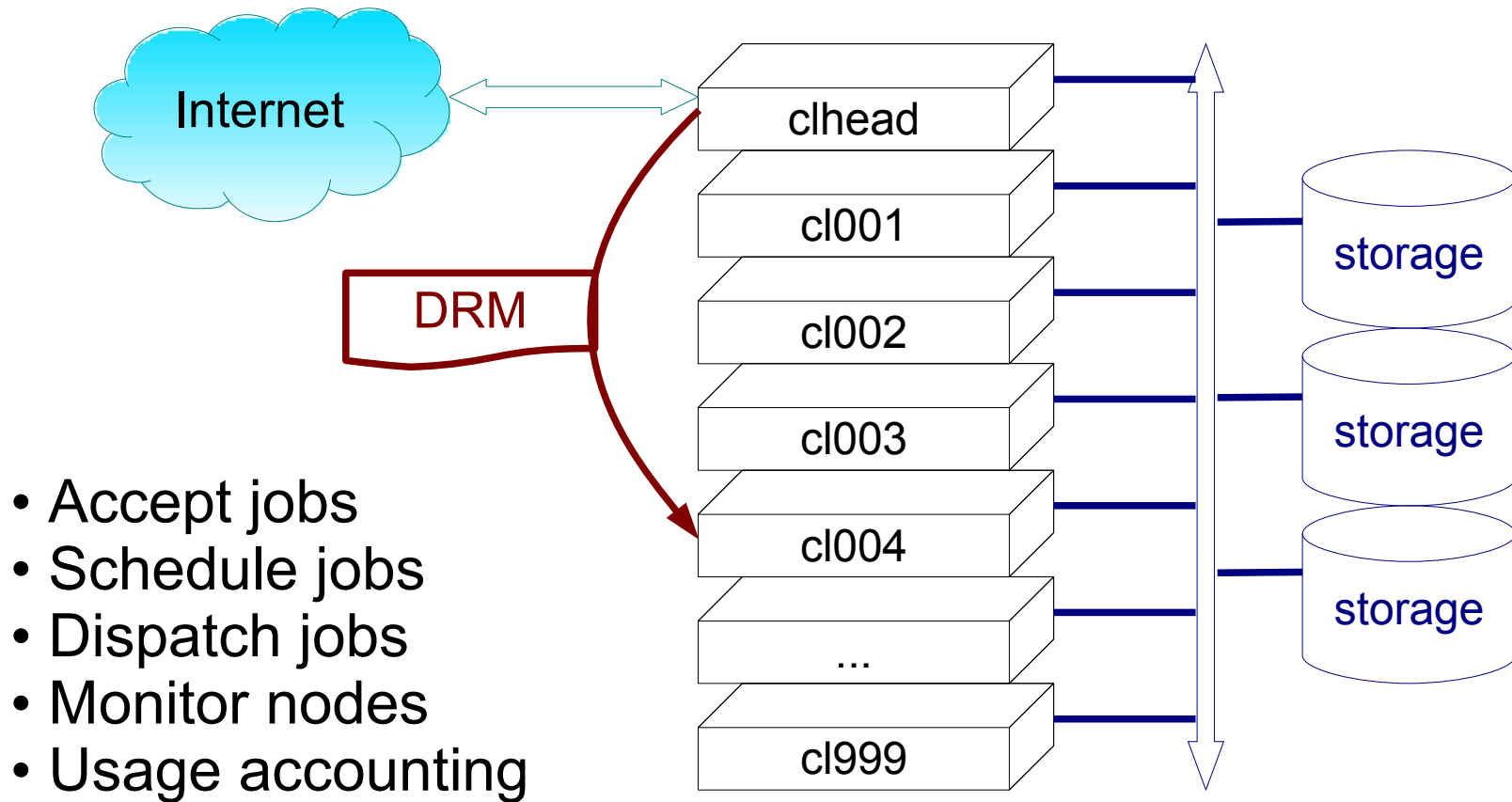
int main (int argc, char *argv[])
{
    int rank, size;
    MPI_Init (&argc, &argv);
    MPI_Comm_rank (MPI_COMM_WORLD, &rank);
    MPI_Comm_size (MPI_COMM_WORLD, &size);
    printf( "This is process %d of %d.\n",
           rank, size );
    MPI_Finalize();
    return 0;
}
```

MPI 'Hello, world'

```
> mpicc mpihello.c -o mpihello  
> mpirun -np 2 mpihello  
This is process 0 of 2.  
This is process 1 of 2.  
>
```

- Testing on head node restricted to <15 CPU mins
- Longer jobs must use dynamic resource manager "Grid Engine"

Role of Dynamic Resource Manager



Dynamic Resource Managers



- Grid Engine used at ACEnet
 - *aka* Sun Grid Engine (SGE), N1GE, or Oracle Grid Engine
- Other DRMs and/or schedulers you might encounter:
 - OpenPBS, Torque, Moab, Maui, LoadLeveler, LSF, SLURM, ...
 - sometimes scheduler is distinct, sometimes bundled

Grid Engine takes a script



```
> cat parscript
#$ -cwd
#$ -l h_rt=0:10:0,test=true
#$ -pe omp* 4
mpirun mpihello
> qsub parscript
Your job 5691 ("parscript") has been
submitted
> qstat
...
```

Grid Engine commands

- **qsub** [*options*] *script*
 - submit a job
- **qstat**
 - is my job running? done?
 - job vanishes from **qstat** when finished
- **qsum**^{*}
 - how busy is the cluster?
- **showq**^{*}
 - how far am I from the head of the waiting list?

^{*} Local scripts, no man pages, no options --- see www.ace-net.ca

qsub flags $\approx$ script directives



- Directives in script...

```
#$ -cwd  
#$ -l h_rt=1:0:0  
./myprog <file.in >file.out
```

- ...equivalent to arguments to qsub

```
qsub -cwd -l h_rt=1:0:0 script
```

- Command line trumps script
- Right-most (last) takes precedence

Job parameters

- **-l h_rt=*hh:mm:ss***
 - hard limit on run time
 - **REQUIRED** - allows scheduler to plan ahead
- **-l test=true**
 - (optional) run on high-availability test nodes
 - must have **h_rt** $\leq$ 1:0:0
- **-l h_vmem=*memory***
 - memory *per parallel slot*
- **-cwd**
 - run in current working directory

Parallel environments

- `-pe parallel_env n_procs`
 - `parallel_env` one of
 - `ompi*`
 - `openmp`
 - `1per, 2per, 4per, 8per, 16per`
 - `n_procs` can be
 - a number,
 - a range, e.g. 4-8
 - or a list, e.g. 4,8,16

Basic scheduling algorithm

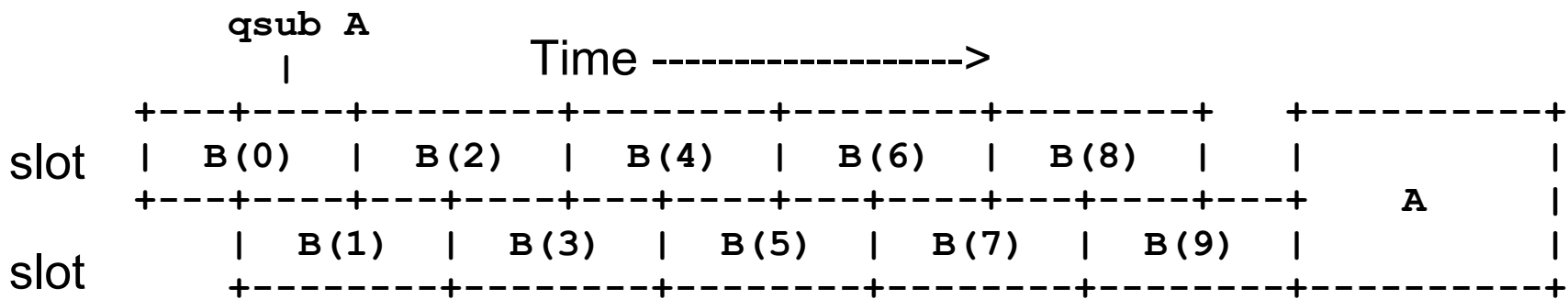


- Don't interrupt running jobs
- Assign priorities to waiting jobs^{*}
- When resources come free:
 - Run highest-priority job that can be satisfied by available resources
 - Iterate until all resources are committed or all jobs have been checked
- Wait for new resources or new jobs

^{*} http://www.ace-net.ca/wiki/Scheduling_Policies_and_Mechanics

Parallel job starvation

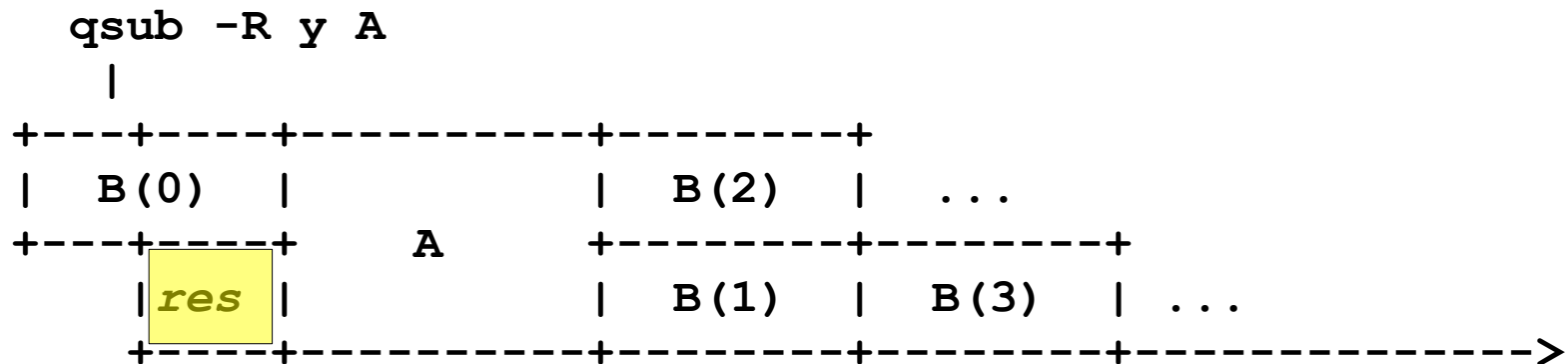
- Jobs scheduled *when resources are available*
- Scheduler tries to fill *all available resources*
- Small jobs (B) can starve “wide” jobs (A) if the small jobs never stop coming:



Solution: Reservation

```
qsub -R y jobscript
```

causes scheduler to **reserve** resources for highest-priority jobs.



...but it's costly, so you have to ask explicitly.
Memory can be the scarce instead of cores.

Some more Grid Engine commands



- **qdel *jobid***
- **qacct *jobid***
 - accounting record, may contain failure hints
- **qhost**
 - host statuses, memory, *etc.*
- **qrsh**
 - interactive session on compute host(s)
- **memslots *memory_amount***
 - local script: See if memory is limiting resource

TotalView debugger

- Compile with `-g` to get symbols (same as any other debugger)
- Ensure X11 working (try `xclock`)
- Use `qrsh` to get interactive session from Grid Engine
- See detailed instructions at <http://www.ace-net.ca/wiki/TotalView>

Where to go for help

- ACEnet User Guide
 - http://www.ace-net.ca/wiki/User_Guide
 - http://www.ace-net.ca/wiki/Job_Control
and links therein
 - http://www.ace-net.ca/wiki/Open_MPI
- User support
 - email: support@ace-net.ca